PTPC: Prototype-Based Test-Time Prediction Calibration for Few-Shot Object Detection

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Abstract

Few-shot detectors can produce plausible post-NMS boxes whose confidence omits two relative cues: competition among novel classes and evidence that a base class better explains the region. We ask whether these omissions can be repaired at inference while keeping detector weights, boxes, and labels fixed. PTPC comprises two composable modules: a contextual novel-prototype rescorer, which pools frozen foundation-model patch tokens and compares the predicted label with all novel support prototypes, and an independently usable Base-Confusability Guard, which continuously attenuates scores according to excess base-versus-novel prototype evidence. Using identical fixed detector outputs, PTPC improves mean VOC AP50 by 11.8 and 5.6 points for DeFRCN and MFDC and COCO AP by 2.4 and 2.5 points; all 42 canonical seed-0 detector–shot cells improve over paired detector-only baselines. For native FSOD-VFM and No-Time-to-Train, every Guard parameter is selected on the other benchmark and then frozen; target-task novel labels appear only in the K supports. The Guard adds 1.91 VOC AP50 and 0.60 COCO AP on FSOD-VFM, and 1.58 VOC AP50/0.75 COCO AP on our paired No-Time reproduction; AP, AP50, ECE, and Brier improve over the native hosts in all 38 cells. A controlled factorial rules out an encoder swap, whereas an absolute-similarity control stays at native AP50. An exact fixed-output ceiling separates remaining ranking headroom from errors that score-only calibration cannot repair.

1. Introduction

Few-shot object detection (FSOD) adapts a detector to novel categories from few labeled supports, which can leave confidence unreliable. After post-NMS boxes and labels are fixed, confidence controls AP ordering and threshold decisions. Monotone calibrators change scale but not within-class order [10, 15], while detector retraining changes the outputs. We instead change only scores, enabling paired ranking and reliability evaluation and an exact label-preserving ceiling. Such rescoring cannot recover missed or mislocalized objects or wrong labels.

The closest post-hoc method, DeFRCN's Prototype Calibration Block (PCB) [25], fuses detector confidence with one *unary* similarity $\cos(q, \mu_{\hat{c}})$ from an independently encoded crop. This score lacks a reference when another novel or base prototype fits better, and crop encoding discards image context. We therefore test whether relative prototype evidence complements the native score beyond an encoder swap.

We introduce Prototype-Based Test-Time Prediction Calibration (PTPC), with two composable modules. The *contextual novel-prototype module* pools patch tokens within each box and compares the predicted label against all novel prototypes. The independently usable *Base-Confusability Guard* continuously attenuates detections better explained by cached base prototypes. Full PTPC composes both; compatible foundation-model detectors can use the Guard alone (Fig. 1). A binary-limit ablation isolates the continuous margin, and final-score temperature scaling evaluates reliability without changing AP order.

Contributions.

1. We combine contextual novel-class competition with base-versus-novel evidence in full PTPC and a native-descriptor Guard interface.
2. On fixed outputs, PTPC gains 11.8/5.6 VOC AP50 and 2.4/2.5 COCO AP for DeFRCN/MFDC, improving all 42 cells. Cross-dataset-frozen Guards gain 1.91/0.60 AP50/AP on FSOD-VFM and 1.58/0.75 on No-Time, with AP, AP50, ECE, and Brier improving in all 38 host cells.
3. Factorial, absolute-only, calibration, and exact-ceiling controls isolate the mechanism and delimit label-preserving repair.

2. Related Work

Nearest prototype and test-time methods. Most FSOD methods alter training through support-conditioned meta-learning or base-to-novel fine-tuning [1, 4, 25, 28, 29, 36]; recent systems further change data generation or objectness learning [21, 38]. Our direct fixed-output precursor is DeFRCN's Prototype Calibration Block (PCB) [25], which scores detector RoIs against offline support prototypes and fuses the assigned-class similarity with the detector confidence. Despite its name, Model Calibration [5] is different: it modifies backbone, RPN, and detector training and mixes base and novel feature distributions. Other nearby methods relax different constraints. TIDE [18] instantiates a support-guided dynamic classifier and cross-attention detector; PS-TTL [7] updates a mean-teacher detector on query images using prototype-derived soft labels; and PDA [12] learns a projection and optional prototype-conditioned RoI alignment during fine-tuning. PTPC instead accepts a completed detection set and updates neither model parameters nor its boxes or labels.

Prototype and foundation-model FSOD. Prototype classifiers originate in metric few-shot learning [27] and appear inside trained FSOD pipelines as support aggregation, class conditioning, or prototype refinement [1, 16, 20, 31, 34]. Such class means are point estimates: with very few supports, a binary nearest- prototype decision can change abruptly near a similarity boundary. Our method keeps the standard mean prototype but uses a source-selected, cross-dataset-frozen Guard; the binary limit is retained as a paired control. Low absolute novel similarity is also not equivalent to positive base evidence; a separate auxiliary-label negative control tests that question without entering any strict-K-shot comparison. Foundation-model pipelines instead build detections from frozen visual or vision–language representations: DE-ViT uses background prototypes [39], FM-FSOD and VisTex-OVLM contextualize image exemplars [11, 33], and reference-based or graph-diffusion systems compose proposal, segmentation, and correspondence modules [2, 6]. Because these pipelines generate their own proposal and label populations, their absolute results provide detector-level context. Our native transfer instead holds each host's outputs fixed and applies only the Guard through its existing region descriptor.

Detection confidence calibration. Classification calibration [10] does not capture the coupling of confidence, localization, and selection in detection. Detection-specific work therefore studies multivariate D-ECE [14], localization-aware calibration [22], and consistent density-based estimation [24]. Conditional post-hoc methods use box size [8] or predicted IoU and duplication structure [9] to improve reliability and sometimes ranking, while Kuzucu et al. [15] show that the evaluation population, matching rule, threshold, and metric can reverse method rankings. These methods estimate reliability from geometric or held-out detection evidence; they do not compare a detection with competing support classes. PTPC is therefore a support-conditioned semantic rescorer, not a conventional scalar calibrator: it combines predicted-label compatibility under all-novel competition with a continuous base-versus-novel gate on a fixed post-NMS population. The two roles are composable rather than interchangeable: we also fit a monotone temperature to the *final* Guard score, which can improve probability reliability but, by construction, cannot change AP ordering.

3. Method

3.1. Problem Setting and Score-Rescoring Goal

Following the standard FSOD setup [4, 13, 25, 29, 30, 34], categories are partitioned into a base set C_b with abundant annotations and a novel set C_n with K labeled support examples per class, with corresponding data D_{base} and D_{novel}; the detector is trained on the former and adapted to the latter, then evaluated on a query set. Our point of departure, PCB [25], encodes each detection and support crop with a frozen ImageNet ResNet-101 and linearly fuses an isolated cosine against the predicted-class prototype with the detector confidence. PTPC separates the two missing relative comparisons into independently usable modules: contextual competition within C_n (§3.3) and a base-versus-novel membership test (§3.4). Full PTPC composes them in that order; a host that already exposes a compatible region descriptor may use only the Guard.

Let D be a frozen few-shot detector. For a query image $I \in [0,1]^{H_I \times W_I \times 3}$ of height H_I and width W_I, D returns a finite post-NMS detection set

$$\mathcal{Y}_D(I) = \{(b_i, c_i, s_i^d)\}_{i=1}^{N_I}, \tag{1}$$

where b_i is an axis-aligned pixel-coordinate box, $c_i \in C_b \cup C_n$ the detector-predicted class, and $s_i^d \in [0,1]$ its native score. Both modules are the identity on base-labelled outputs. For $c_i \in C_n$, let g_N denote contextual novel-prototype rescoring and g_G the Guard. The supported compositions are

$$\hat{s}_i = \begin{cases} s_i^d, & c_i \in C_b, \\ g_G(g_N(s_i^d)), & \text{full PTPC}, \\ g_N(s_i^d), & \text{novel module only}, \\ g_G(s_i^d), & \text{native-descriptor Guard only}. \end{cases} \tag{2}$$

Here the shorthand suppresses dependence on I, b_i, c_i and the support sets. The modules never alter a box, label, or detection identity. Their raw outputs are semantic ranking scores, not asserted estimates of $P(\text{TP} \mid I, b_i, c_i)$; probability calibration is evaluated separately in §4.1.

3.2. Frozen Encoder for Full PTPC

PTPC uses a frozen foundation-model encoder ϕ whose full-image forward pass exposes a spatial grid of patch tokens. We take the normalized final-block spatial tokens, exclude prefix tokens, and apply each checkpoint's native color conversion and channel normalization. Let S_ϕ denote the encoder's square input resolution and let $R(I)$ include this preprocessing and the deterministic square resize used by the implementation. The encoder maps the resized image to a grid of patch tokens,

$$\phi : \mathbb{R}^{S_\phi \times S_\phi \times 3} \to \mathbb{R}^{H \times W \times d}, \qquad Z = \phi(R(I)), \tag{3}$$

where H, W are the token-grid height and width, d the embedding dimension, and $Z_{u,v} \in \mathbb{R}^d$ the token at grid row u and column v. Flattening the grid gives $T(I) = \{(p_r, t_r)\}_{r=1}^{HW}$ with $t_r = Z_{u,v}$ at flat index $r = (u-1)W+v$ and patch-center coordinate $p_r = (x_r, y_r) \in [0, S_\phi]^2$ given by

$$x_r = \frac{v - \frac{1}{2}}{W} S_\phi, \qquad y_r = \frac{u - \frac{1}{2}}{H} S_\phi, \tag{4}$$

placing each token at the center of its grid cell.

3.3. Contextual Novel-Prototype Module

Token selection inside each detector box. Let Π be the deterministic rescaling map from the original image frame to the $S_\phi \times S_\phi$ encoder frame:

$$\begin{aligned} \tilde{b}_i &= \Pi(b_i) = (\tilde{x}_i^1, \tilde{y}_i^1, \tilde{x}_i^2, \tilde{y}_i^2), \\ \tilde{x}_i^j &= S_\phi \frac{x_i^j}{W_I}, \qquad \tilde{y}_i^j = S_\phi \frac{y_i^j}{H_I}, \qquad j \in \{1, 2\}, \end{aligned} \tag{5}$$

with center $p_{b_i} = (x_{b_i}, y_{b_i}) = \big(\frac{1}{2}(\tilde{x}_i^1 + \tilde{x}_i^2),\ \frac{1}{2}(\tilde{y}_i^1 + \tilde{y}_i^2)\big)$. Using the token centers $p_r = (x_r, y_r)$ from Eq. (4), let $\Omega_{\text{in}}(b_i) = \{r \mid \tilde{x}_i^1 \le x_r \le \tilde{x}_i^2,\ \tilde{y}_i^1 \le y_r \le \tilde{y}_i^2\}$ collect the token indices $r \in \{1, \dots, HW\}$ whose centers fall inside the rescaled box. PTPC then selects

$$\Omega(b_i) = \begin{cases} \Omega_{\text{in}}(b_i), & \Omega_{\text{in}}(b_i) \neq \emptyset, \\ \big\{\arg\min_r \|p_r - p_{b_i}\|_2\big\}, & \text{otherwise}, \end{cases} \tag{6}$$

falling back to the nearest token whenever no center lies inside the rescaled box so that $\Omega(b_i)$ is never empty.

Center-weighted RoI pooling. Selected tokens are pooled with Gaussian weights centred on the box, so that evidence near the object centre dominates evidence near the box edge. With w_i, h_i the box's encoder-frame width and height and δ the token stride ($\delta = \min(S_\phi/W, S_\phi/H)$ for a regular grid), the bandwidths are $\sigma_{x,i} = \max(w_i/2, \delta)$ and $\sigma_{y,i} = \max(h_i/2, \delta)$: the pooling footprint tracks object scale, while the stride floor prevents collapse for tiny boxes. Writing $d_{r,i} = \big((x_r - x_{b_i})/\sigma_{x,i}\big)^2 + \big((y_r - y_{b_i})/\sigma_{y,i}\big)^2$ and $\text{norm}(z) = z/\|z\|_2$, the patch-token RoI descriptor is

$$\begin{aligned} q_p(I, b_i) &= \text{norm}\Big(\textstyle\sum_{r \in \Omega(b_i)} a_r(b_i)\ \text{norm}(t_r)\Big), \\ a_r(b_i) &= e^{-d_{r,i}/2} \Big/ \textstyle\sum_{u \in \Omega(b_i)} e^{-d_{u,i}/2}. \end{aligned} \tag{7}$$

Unlike an independently encoded crop-CLS embedding, each selected token was computed in full-image context. The pooling rule localizes the readout to the detector box and mildly downweights boundary content; it does not remove information propagated from outside the box by self-attention.

Novel patch supports and prototypes. For each novel class $c \in C_n$ with K-shot support set $\mathcal{S}_c = \{(I_c^j, b_c^j)\}_{j=1}^K$, applying the same extractor (Eqs. (5)–(7)) to every support pair and averaging over the K shots gives the class patch prototype $\mu_c^p = \text{norm}\big(\frac{1}{K}\sum_j q_p(I_c^j, b_c^j)\big)$.

Competitive compatibility over novel classes. Given a query detection (b_i, c_i), PTPC compares the patch-level query descriptor with all novel patch prototypes by competitive scoring over the novel label space:

$$s_p(b_i, c) = \frac{\exp\big(q_p(I, b_i)^\top \mu_c^p / \tau\big)}{\sum_{c' \in C_n} \exp(q_p(I, b_i)^\top \mu_{c'}^p / \tau)}. \tag{8}$$

Each inner product is exactly cosine similarity since $q_p(I, b_i)$ and μ_c^p are unit-norm by construction (Eq. (7) and the prototype definition above). The softmax temperature $\tau \in (0, \infty)$ controls competition sharpness. This is a conditional compatibility score over the supplied novel label set, not a probability that the box is a correct detection: background and base classes are absent from the denominator, and its scale depends on $|C_n|$. PTPC evaluates Eq. (8) at the detector-predicted label c_i, giving the novel-module score $s_p(b_i, c_i)$.

Fusion with detector confidence. PTPC fuses this novel-prototype evidence with the original detector confidence:

$$s_{\text{fuse}}(b_i) = w_d\, s_i^d + w_\phi\, s_p(b_i, c_i), \qquad w_d + w_\phi = 1, \tag{9}$$

with $w_d, w_\phi \in [0, 1]$: the frozen-encoder branch is a corrective signal driven by the novel support set, not a replacement detector. Following the common PCB evaluation interface, the novel module evaluates Eq. (9) only where the native score satisfies $\ell_N < s_i^d \le u_N$; outside this band we set $s_{\text{fuse}}(b_i) = s_i^d$. Eligibility is an explicit Boolean mask on the native score. The Guard has its own band and independently recomputes eligibility on its input score, because the novel module can move a detection across a boundary. The deployed full model happens to use $(\ell_N, u_N] = (\ell_G, u_G] = (.05, 1]$, but the modules are not mathematically coupled to a shared band.

3.4. Base-Confusability Guard Module

Competitive scoring over novel prototypes alone still gives high confidence to a base-class object the detector has labelled novel, so full PTPC applies the Guard after Eq. (9). The Guard can also operate independently. It encodes each detection and support example as one normalized region descriptor and averages the support descriptors into one vector per base class. In full PTPC this is a global crop descriptor on the same frozen encoder. Crop-CLS is not assumed necessary: §4.3 shows that the novel module's patch-RoI descriptor performs equivalently. It removes crop-CLS encoding and admits patch-grid cache sharing, although the released accuracy path re-extracts the descriptor for the Guard's independently eligible subset.

Descriptor-compatible Guard interface. Some foundation-model detectors already compute a region feature $h_D(I, b_i)$ for every final detection. In that case we set $q_g^D(I, b_i) = \mathrm{norm}(h_D(I, b_i))$. The host adapter must expose the same semantic feature space for annotated support regions; it then forms base and novel prototypes with the identical averaging rule below. Query and support localization mechanisms need not be identical—for example, a host may expose mask-pooled query features but only box-derived support masks—and each such choice is reported. The Guard is therefore *descriptor-interface compatible*, not universally architecture-agnostic: it requires final detections, matched support/query descriptors, and representative support.

Crop-level guard descriptor. Let $\mathrm{crop}(I, b_i) \in [0,1]^{S\times S\times 3}$ be the region inside b_i, cropped from I and resized to the encoder crop resolution $S \times S$, and let ϕ_g be the frozen encoder's native global crop representation: its CLS embedding when a prefix token is defined, and its pre-logit global pool otherwise (as for prefix-free SigLIP models). It shares weights with ϕ and differs only in input (a crop at resolution S) and output (one global embedding instead of the spatial patch grid). The crop-level descriptor is $q_g(I, b_i) = \mathrm{norm}(\phi_g(\mathrm{crop}(I, b_i))) \in \mathbb{R}^d$. For compactness, write $\bar{q}_g = q_g$ for full PTPC and $\bar{q}_g = q_g^D$ for a native-host adapter. Within either use, query and support regions are mapped into the same descriptor space.

Novel and cached base region prototypes. Applying $\bar{q}_g$ to the novel support set $\mathcal{S}_c$ and averaging over the K shots yields the novel region prototype μ_c^g; the same descriptor applied to M_k retained crops of each base class $k \in C_b$ yields a cached base prototype ν_k^g, used only as negative evidence:

$$\begin{aligned} \mu_c^g &= \mathrm{norm}\Big(\tfrac{1}{K}\textstyle\sum_j \bar{q}_g(I_c^j, b_c^j)\Big), \\ \nu_k^g &= \mathrm{norm}\Big(\tfrac{1}{M_k}\textstyle\sum_m \bar{q}_g(I_k^m, b_k^m)\Big). \end{aligned} \tag{10}$$

Multiplicative score suppression. The Guard compares $\bar{q}_g(I, b_i)$ against the nearest novel and nearest base region prototype, $\delta_i^+ = \max_{c\in C_n} \bar{q}_g(I, b_i)^\top \mu_c^g$ and $\delta_i^- = \max_{k\in C_b} \bar{q}_g(I, b_i)^\top \nu_k^g$. The maximum over C_n deliberately asks whether the region resembles the *novel set*; it does not certify the fixed label c_i. Label-specific novel competition is the role of g_N; the supplementary reference-set audit separately tests using only $\mu_{c_i}^g$ as the positive reference. Because maxima depend on the number of candidates, the same analysis also varies the number of retained base classes; changing crops per prototype M_k is a distinct uncertainty axis.

The Guard acts only inside its score band $(\ell_G, u_G]$. Eligibility is computed by an explicit Boolean mask on s_i^{in}, not by slicing a presumed score-sorted prefix. In full PTPC this mask is recomputed after the novel module. To avoid an abrupt decision at a noisy few-shot similarity boundary, the canonical Guard maps excess base evidence continuously to a multiplicative gate. With σ the logistic sigmoid and margin temperature $\eta > 0$,

$$\begin{aligned} B_i &= \mathbf{1}[\ell_G < s_i^{\mathrm{in}} \le u_G], \\ h_i^{(\eta)} &= B_i\, \sigma\!\left(\frac{\delta_i^- - \delta_i^+ - m}{\eta}\right), \\ g_i^{(\eta)} &= 1 - (1-\gamma) h_i^{(\eta)}, \\ s_{\mathrm{final}}(b_i) &= g_i^{(\eta)} s_i^{\mathrm{in}}, \\ s_i^{\mathrm{in}} &= \begin{cases} s_{\mathrm{fuse}}(b_i), & \text{full PTPC}, \\ s_i^d, & \text{native Guard-only}. \end{cases} \end{aligned} \tag{11}$$

where $0 \le \ell_G < u_G \le 1$ and $\gamma \in (0, 1]$. Far below the margin the multiplier approaches one; far above it the multiplier approaches γ. Ambiguous regions near the boundary therefore do not receive an all-or-nothing penalty. The binary-limit control replaces $\sigma((\delta_i^- - \delta_i^+ - m)/\eta)$ by $\mathbf{1}[\delta_i^- - \delta_i^+ > m]$; for every non-boundary detection it is the $\eta \downarrow 0$ limit of Eq. (11). Neither rule relabels, deletes, or moves a detection. Boxes, labels, stored output order, and count are therefore invariant; only score-induced ranking and threshold membership can change. Unless explicitly marked *binary limit*, "Guard" and "PTPC" refer to the continuous rule in Eq. (11) throughout the paper.

Caching and query-time cost. Support prototypes are computed once. The deployed full crop-CLS path caches $(2|C_n| + |C_b|)d$ scalars; Guard-only or a patch-RoI composition that shares the novel supports caches $(|C_n| + |C_b|)d$.

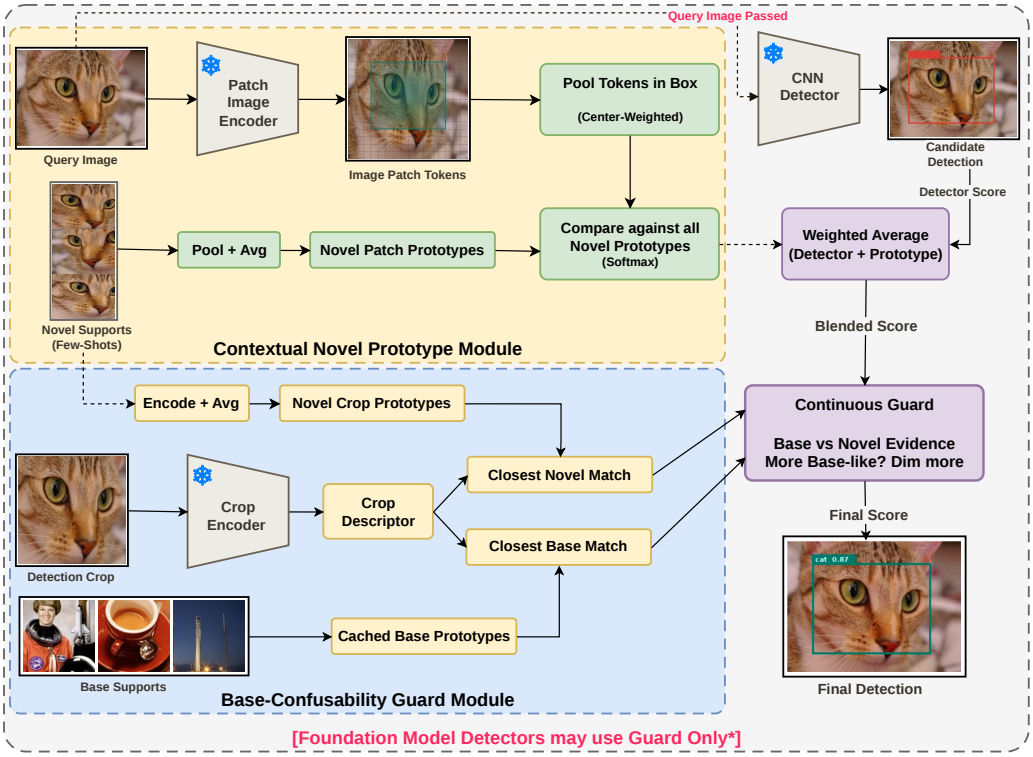


Figure 1. **Composable PTPC modules.** The upper branch competitively fuses contextual novel-prototype evidence; the independently usable Guard continuously attenuates scores under excess base evidence. Full PTPC composes both, while compatible native hosts may use the Guard alone. Boxes and labels remain fixed.

Query inference uses one full-image pass for the novel module; a crop-CLS Guard additionally encodes its eligible crops. Native Guard-only transfer instead reuses host descriptors and adds only prototype-similarity matrix products, with no new query encoder pass. Exact complexity, cache accounting, latency, and hardware are supplementary.

4. Experiments

4.1. Experimental Setup

Benchmarks and paired protocol. We evaluate full PTPC with DeFRCN [25] and MFDC [32] on PASCAL VOC [3] and MS COCO [19] under the SNIDA protocol [30]; FSCE [28] is a supplementary detector-family control. Guard-only transfer uses FSOD-VFM [6] and No-Time-to-Train [2]. Our CNN rows use the authors' code and canonical seed-0 supports. Every paired comparison shares the checkpoint, support, post-NMS boxes, labels, count, and order; only scores change. Published rows are therefore context, not paired controls.

Frozen configuration and label accounting. Full PTPC uses DINOv2-B/14, $(w_d, w_\phi, \tau) = (.4, .6, .1)$, $(m, \gamma) = (.05, .3)$, score bands $(.05, 1]$, and $M = 20$ cached base crops per class. Its non-η tuple is selected once on labelled VOC Split 1 development data and then frozen across detectors, other splits, and COCO. For native-host transfer, *every* Guard parameter is selected on the other benchmark; target labels only form the normal K-shot prototypes. Auxiliary diagnostics use a test-disjoint 1,000-image labelled population and are excluded from strict-K-shot claims. No query-test label selects a rule. Complete tuples and grids, preprocessing, label exclusions, identity hashes, and probability-map fitting are specified in the supplement.

Metrics. We report AP, AP50, and AP75. On the fixed detection population, same-class one-to-one matching at IoU$\geq .5$ defines correctness for 15-bin ECE [10] and Brier score. Class-balanced LaECE$_0$ [15, 22], threshold sweeps, calibration baselines, and full definitions are supplementary; monotone probability maps are separated from AP because they cannot change ranking.

4.2. Main Results

The two main comparisons use canonical continuous-Guard PTPC-B and exact paired outputs; binary-rule, encoder-scale, and mechanism controls are treated as ablations below or in the supplement.

PASCAL VOC. On PASCAL VOC (Table 1), canonical PTPC-B improves the identical DeFRCN outputs from 49.6 to 61.4 mean AP50 and MFDC from 57.7 to 63.3, improving all 30 paired detector–shot cells. Gains persist at 10-shot, so they are not explained only by stabilizing a one-shot prototype; the smaller MFDC gain instead tracks its lower remaining ranking headroom. With all Guard parameters selected on COCO, FSOD-VFM rises 78.2→80.1 and No Time rises 68.1→69.7 mean AP50, each improving all 15 cells without changing boxes or labels. Published retraining and proposal-replacement methods remain useful context, but are not paired fixed-output controls.

MS COCO. On MS COCO (Table 2), canonical PTPC-B improves its paired DeFRCN detector from 15.7 to 18.1 mean AP ($+2.4$) and MFDC from 16.4 to 18.8 ($+2.5$), improving all 12 paired cells. The smaller one-shot gains are consistent with noisier novel prototypes, yet the improvement remains broadly stable from two shots onward rather than vanishing with K. Under VOC-only selection, FSOD-VFM rises 39.8→40.4 mean AP; No Time rises 36.3→37.1 at 10 shots and 37.0→37.7 at 30 shots. Thus the same relative scoring cues transfer across datasets and native hosts, while methods that replace the output set answer a complementary question.

4.3. Ablations and Scope

Continuous Guard and rule selection. On fixed FSOD-VFM outputs, the continuous Guard exceeds its binary limit by 0.28 VOC AP50 and 0.11 COCO AP, wins 14/18 AP/AP50 comparisons across 21 cells, and always exceeds native. Source-selected $\eta = .04/.01$ avoids target tuning. The direction persists for DeFRCN/MFDC (VOC +0.672/+0.596; COCO +0.0816/+0.0898) and No-Time (+0.39 VOC AP50, +0.04 COCO AP). Thus continuity modestly refines the main gain; all four metrics improve in all 38 source-frozen host cells.

Mechanism controls. Absolute-only rules remain at native AP50 (78.19/78.26 versus 78.23), while the relative Guard reaches 79.75. Temperature fitting reduces native ECE from 0.150 to 0.040 and Guard ECE/Brier to 0.0117/0.0064 without changing AP. These test-disjoint 1,000-image diagnostics isolate mechanism and reliability, not strict-K-shot gains; full tables are supplementary.

Encoder robustness. Table 3 varies only the frozen foundation encoder on fixed MFDC Split 1 outputs. All eight encoders exceed the detector baseline, including CLIP and SigLIP, while DINOv2-L is strongest on this split. The non-monotonic DINOv2 scale trend argues against capacity alone explaining the gain; all-split VOC and COCO sweeps are supplementary.

Robustness and deployment controls. Support budget saturates: $M = 5/20/50$ gives 65.42/66.18/66.34 AP50, whereas base coverage $B = 5/10/15$ gives 63.33/64.41/66.18. Gains persist across DeFRCN retrainings ($+11.05 \pm 0.09$; 15/15), VOC support draws for two hosts ($+3.4$–$+13.6$; 120/120), and MFDC COCO runs ($+2.5 \pm 0.4$ AP; 4/4). Detector/novel/full inference takes 48.5/78.4/112.4 ms per image; further controls are supplementary.

Factorial component reasoning. The fixed-output factorial (Table 4) separates encoder, pooling, competition, and Guard. DINOv2 improves crop-CLS AP50 (56.09→60.68) but worsens ECE (0.217→0.369), rejecting an encoder-only explanation. Competition plus Guard loses 1.62 AP50 with global crops but gains 2.15 with localized patch-RoI evidence; the complete patch configuration is best in mean AP50 and Brier.

Fixed-output ceiling. At 10-shot, PTPC (21.69 AP) remains 14.87 points below the fixed-label bound (Fig. 3). Yet the 10/30-shot bounds (36.56/40.35) remain below output-changing FSOD-VFM (44.0/45.8) [6] and VisTex (52.7/53.6) [33]: rescoring has ranking headroom but cannot repair the detection population.

Guard selectivity. Across 533,937 checks, the crop Guard suppresses 79.9% without novel-GT overlap versus 10.6% with overlap (patch-RoI: 90.7%/9.8%), rejecting uniform shrinkage while exposing recall risk.

Failure and operating-point boundary. Of 40,583 changed Split 2 5-shot detections, 29,839 are demoted false positives and 9,320 are promoted. At threshold 0.5, however, 2,255 false positives cross upward versus 1,222 downward, so AP and a chosen operating point can disagree. Poor boxes (3,179) and wrong labels (1,139) remain immutable. Increasing margin 0.05→0.15 restores horse recall but costs 0.46 mean AP50.

Hyperparameter sensitivity. One-at-a-time sweeps (Fig. 2) span 2.35 AP50 for τ, 0.73 for m, and 0.42 for γ. Alternatives improve the Split 1-selected mean by at most 0.21; uniform pooling changes it by 0.03.

5. Conclusion

PTPC repairs fixed-output scores through contextual novel competition and base-versus-novel evidence. All 42 paired CNN cells and 38 source-frozen native-host cells improve. Factorial and absolute-only controls reject an encoder swap or low-similarity threshold as the explanation, while repeated runs reject a favorable checkpoint or support draw.

Table 1. **PASCAL VOC novel AP50.** Published rows retain source protocols; our rows are exactly paired on seed-0 outputs. Guard rows rescore the corresponding *our impl.* host with COCO-selected $\eta = .04$ and no extra VOC labels. †Our No-Time reproduction uses box-derived, not offline-SAM, support masks. Binary/encoder controls are supplementary. **Bold**/underlined: best/second best per column.

Method	Novel Split 1					Novel Split 2					Novel Split 3					Mean
	1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
Norm-VAE [35], CVPR 23	62.1	64.9	67.8	69.2	67.5	39.9	46.8	54.4	54.2	53.6	58.2	60.3	61.0	64.0	65.5	59.1
DeFRCN [25], ICCV 21 (our impl.)	48.9	56.3	54.5	60.4	56.1	28.4	39.5	47.5	52.1	47.0	43.7	47.2	51.2	54.6	56.9	49.6
MFDC [32], ECCV 22 (our impl.)	61.7	66.1	60.9	68.8	66.5	41.3	41.1	45.5	54.8	52.9	58.0	60.7	59.1	63.0	64.5	57.7
SNIDA-MFDC [30], CVPR 24	64.9	67.9	69.7	71.4	70.5	42.2	47.8	54.5	56.6	54.9	58.1	61.3	60.7	63.6	66.0	60.7
MPAD [21], ICLR 25	69.1	69.5	69.6	69.9	68.9	58.4	59.7	61.8	61.8	63.5	**70.1**	69.8	69.9	70.4	71.4	66.9
ViT / Foundation-Model methods																
DE-ViT-B [39], CoRL 24	56.9	61.8	68.0	73.9	72.8	45.3	47.3	58.2	59.8	60.6	58.6	62.3	62.7	64.6	67.8	61.4
FM-FSOD-L [11], CVPR 24	40.1	53.5	57.0	68.6	72.0	33.1	36.3	48.8	54.8	64.7	39.2	50.2	55.7	63.4	68.1	53.7
VisTex-GLIP [33], ICCV 25	69.4	71.8	73.4	74.6	74.8	66.5	70.1	73.1	74.2	74.4	68.9	70.2	71.1	72.3	72.6	71.8
No Time to Train [2], arXiv 25	70.8	72.3	73.3	77.2	79.1	54.5	67.0	76.3	75.9	78.2	61.1	67.9	71.3	70.8	72.6	71.2
No Time to Train (our impl.)†	71.9	70.8	64.0	69.9	76.5	54.6	62.9	71.6	72.7	74.1	57.7	65.2	68.4	69.3	71.5	68.1
FSOD-VFM [6], ICLR 26	77.5	82.3	83.0	85.8	85.8	64.8	68.0	77.4	79.5	81.6	65.3	75.1	78.7	78.2	79.3	77.5
(our impl.)	77.9	83.0	84.0	86.6	86.5	65.4	69.2	78.5	79.8	82.2	65.5	75.5	79.7	79.0	80.3	78.2
Ours (DeFRCN + PTPC-B)	64.7	69.7	70.5	73.6	71.8	38.0	46.4	58.0	60.6	60.1	56.3	58.6	61.6	65.1	66.4	61.4
Ours (MFDC + PTPC-B)	70.2	67.4	73.6	75.9	76.4	42.0	43.6	56.0	60.1	60.5	60.8	63.3	64.2	67.2	68.4	63.3
Ours (No Time + Guard)†	73.8	73.3	67.0	73.2	79.4	55.0	63.7	72.8	73.9	75.1	59.0	66.9	69.1	70.3	72.2	69.7
Ours (FSOD-VFM + Guard)	**81.5**	**84.6**	**85.4**	**88.3**	**88.3**	**67.6**	**73.2**	**81.8**	**82.3**	**84.0**	67.3	**76.5**	**80.4**	**79.8**	**81.0**	**80.1**

Table 2. **MS COCO novel AP.** Published protocols are unpaired; our rows share seed-0 outputs. Guard rows use VOC-selected $\eta = .01/.005$ for FSOD-VFM/No Time and no extra COCO labels. †Box-derived-mask No-Time reproduction; lower shots are nested support prefixes. **Bold**/underlined: best/second best per column.

Method	1	2	3	5	10	30	Mean
DeFRCN [25], ICCV 21 (our impl.)	9.3	12.9	14.8	16.1	18.5	22.6	15.7
MFDC [32], ECCV 22 (our impl.)	10.8	13.9	15.0	16.4	19.4	22.7	16.4
SNIDA-MFDC [30], CVPR 24	12.0	15.4	16.4	17.8	20.7	23.8	17.7
OWOBJ-DeFRCN [38], CVPR 25	11.9	15.6	17.9	19.4	23.8	26.4	19.2
MPAD-DeFRCN [21], ICLR 25	18.3	18.5	18.8	18.9	–	–	–
ViT / Foundation-Model methods							
DE-ViT-L [39], CoRL 24	–	–	–	–	34.0	34.0	–
FM-FSOD-L [11], CVPR 24	5.7	11.0	15.7	21.9	27.7	37.0	19.8
VisTex-GLIP [33], ICCV 25	**47.9**	**51.8**	**52.2**	**52.6**	**52.7**	**53.6**	**51.8**
No Time to Train [2], arXiv 25	–	–	–	–	36.6	36.8	–
No Time to Train (our impl.)†	–	–	–	–	36.3	37.0	–
FSOD-VFM [6], ICLR 26	–	–	–	–	44.0	45.8	–
(our impl.)	27.0	35.0	40.4	43.6	45.5	47.3	39.8
AgentDet [17], CVPR 26	10.8	–	16.1	23.9	31.6	37.5	–
Ours (DeFRCN + PTPC-B)	10.5	15.7	17.4	18.6	21.1	25.1	18.1
Ours (MFDC + PTPC-B)	12.5	17.0	17.7	19.6	21.7	24.6	18.8
Ours (No Time + Guard)†	–	–	–	–	37.1	37.7	–
Ours (FSOD-VFM + Guard)	27.1	35.2	41.1	44.6	46.4	48.0	40.4

The continuous Guard improves mean ranking over its binary limit; calibration and ceiling audits delimit score-only repair. Code and frozen artifacts will be released.

Limitations. PTPC cannot fix missed objects, poor boxes, or wrong labels; confusable classes can trade threshold recall for precision. Continuous Guard is not reliability-dominant in every cell. Native transfer needs compatible descriptors and representative base support and is tested only on FSOD-VFM and our box-mask No-Time reproduction. Auxiliary 1,000-image studies are not strict K-shot results, and budget/class-count controls use deterministic subsets. Deployment therefore requires host-specific validation of the Guard and probability map.

Table 3. **Frozen-encoder robustness on MFDC VOC Split 1.** Novel AP50 on fixed outputs; only the encoder changes.

Encoder	1-shot	2-shot	3-shot	5-shot	10-shot	Mean
MFDC (no PTPC)	61.70	66.10	60.90	68.80	66.50	64.80
DINOv2-S/14 [23]	67.27	61.43	71.27	74.88	74.97	69.96
DINOv2-B/14 [23]	68.70	**71.10**	71.10	74.80	73.60	71.86
DINOv2-L/14 [23]	**69.06**	68.37	**72.97**	**76.19**	**76.93**	**72.70**
DINOv2-G/14 [23]	68.45	59.81	70.97	74.78	75.64	69.93
CLIP-B/16 [26]	65.69	59.16	68.06	71.78	71.00	67.14
CLIP-L/14 [26]	64.29	62.99	68.20	74.86	72.94	68.66
SigLIP-B/16 [37]	59.95	63.58	68.29	74.53	75.40	68.35
SigLIP-L/16 [37]	59.44	54.11	66.05	72.67	74.66	65.39

Table 4. **DeFRCN VOC 5-shot factorial ablation.** Rows share fixed outputs and supports; split columns are novel AP50. **Bold**/underlined: best/second best.

Encoder	Pooling	Comp.	Guard	Split 1	Split 2	Split 3	Mean AP50	ECE↓	Brier↓
—	—	—	—	49.98	52.68	50.31	50.99	0.151	0.059
R101	Crop-CLS	—	—	60.59	52.96	54.74	56.09	0.217	0.084
R101	Crop-CLS	✓	—	60.20	48.57	54.59	54.45	0.228	0.103
R101	Crop-CLS	—	✓	61.90	54.01	57.68	57.86	**0.114**	0.044
R101	Crop-CLS	✓	✓	62.04	51.63	57.90	57.19	0.123	0.053
DINOv2-B	Crop-CLS	—	—	65.06	56.54	60.43	60.68	0.369	0.158
DINOv2-B	Crop-CLS	✓	—	68.80	59.51	61.61	63.31	0.205	0.073
DINOv2-B	Crop-CLS	—	✓	70.23	59.76	63.75	64.58	0.166	0.054
DINOv2-B	Crop-CLS	✓	✓	68.94	58.49	61.46	62.96	0.205	0.073
DINOv2-B	Patch-RoI	—	—	65.95	56.77	60.82	61.14	0.381	0.168
DINOv2-B	Patch-RoI	✓	—	68.05	57.26	60.00	61.77	0.240	0.102
DINOv2-B	Patch-RoI	—	✓	72.11	59.05	65.14	65.43	0.173	0.058
DINOv2-B	Patch-RoI	✓	✓	**74.82**	**60.31**	**67.60**	**67.58**	**0.114**	**0.043**

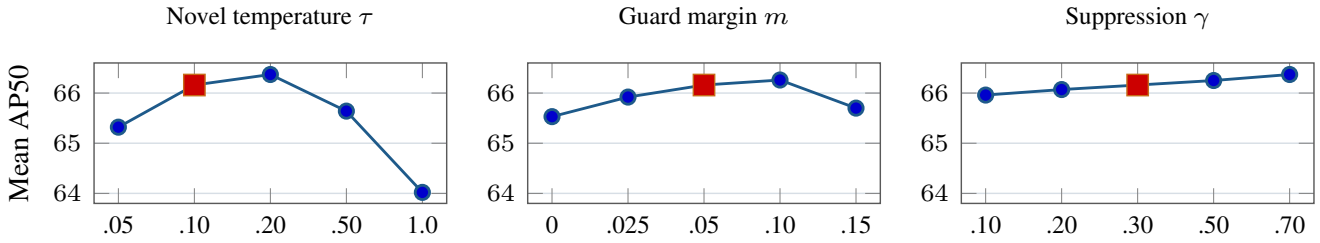


Figure 2. **Hyperparameter sensitivity on DeFRCN VOC 5-shot.** Three-split mean AP50 for one-at-a-time sweeps. Squares mark the Split 1-selected settings; split-wise values are supplementary.

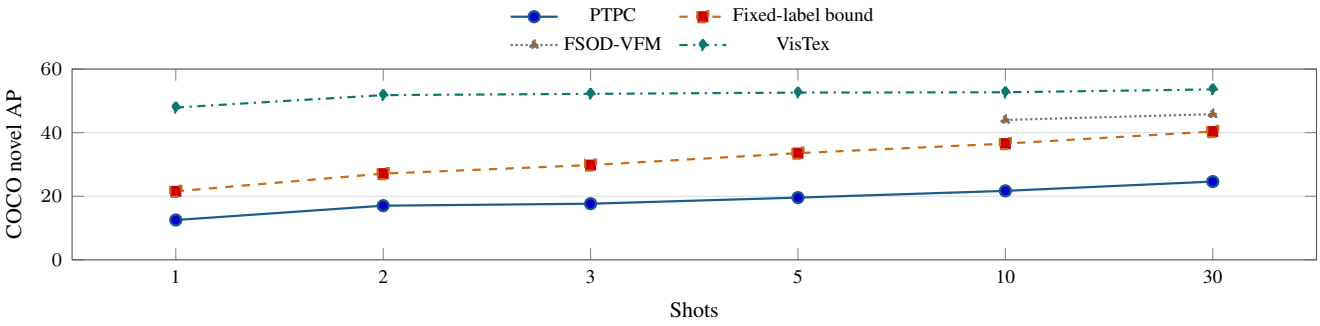


Figure 3. **Fixed-output ceiling.** The label-preserving bound leaves ranking headroom but trails output-changing pipelines. COCO novel AP.

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